

CYBERSECURITY PRACTICAL LAB

Cybersecurity Virtual Lab Setup

Lab Setup Report

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Date	<u>10/09/2026</u>

LAB COMPONENTS

Component	Purpose
Kali Linux	Security testing workstation
Metasploitable 2	Intentionally vulnerable target VM
VMware Workstation	Virtualization platform
Wireshark	Network packet capture and analysis

1. Introduction

This lab establishes an isolated cybersecurity testing environment consisting of Kali Linux and Metasploitable 2. Kali Linux is used as the security testing workstation, while Metasploitable 2 acts as an intentionally vulnerable target system. Wireshark is used to capture and analyze network traffic generated during connectivity tests. The environment provides a controlled platform for learning Linux administration, network reconnaissance, vulnerability assessment, exploitation, and incident analysis.

1.1 Lab Objectives

The objectives are to configure two virtual machines on an isolated network, verify IP connectivity, perform basic service discovery, and capture test traffic using Wireshark.

1.2 Recommended Network Topology

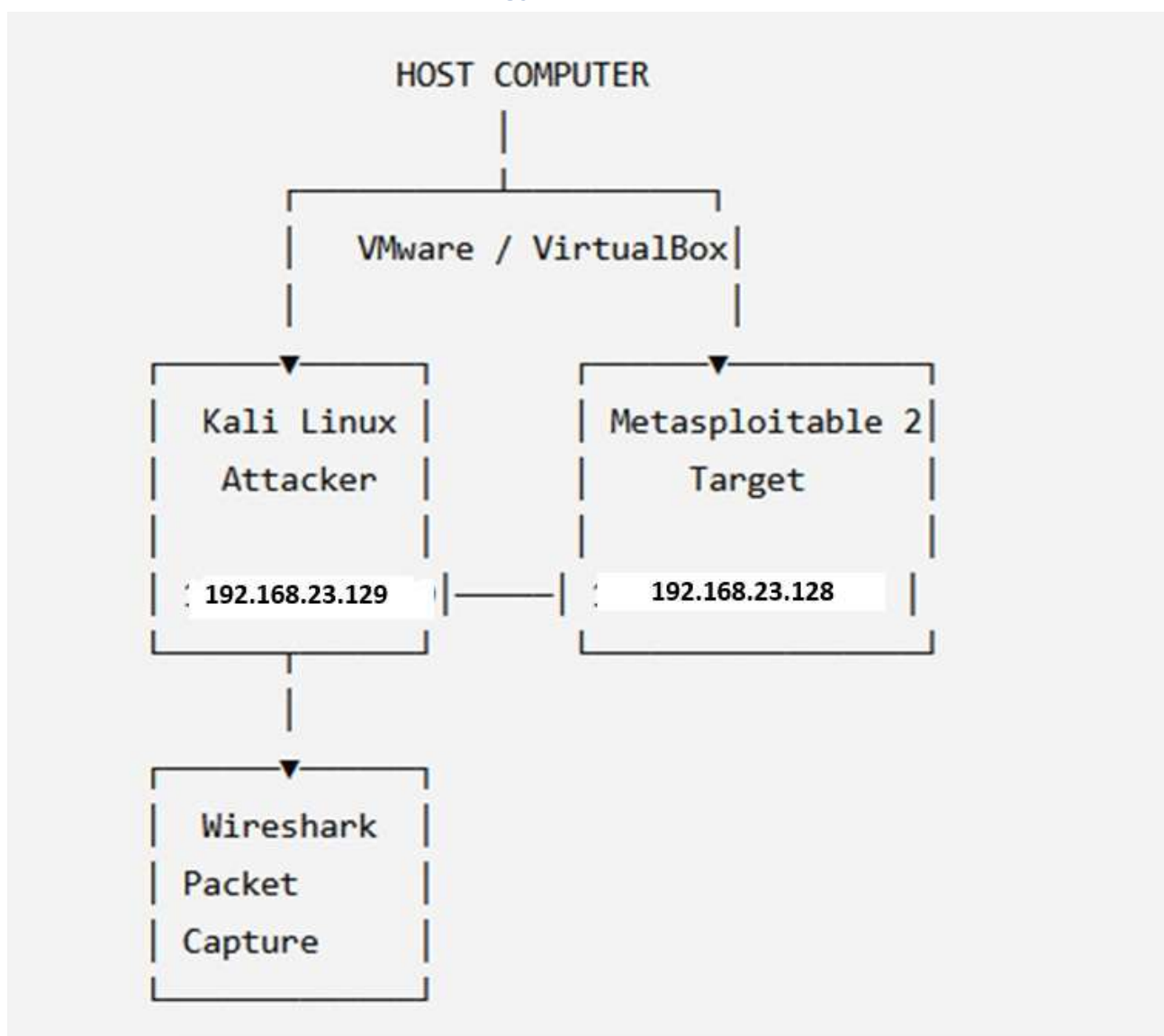


Figure 1 — Isolated lab topology

Safety Note

Metasploitable 2 is intentionally vulnerable. It should remain on an isolated host-only/internal network and should not be exposed directly to an untrusted or production network.

2. Step 1 — Install / Configure Kali Linux

Start the Kali Linux virtual machine and verify the operating system, kernel, current user, and network configuration.

2.1 Verify Operating System

```
cat /etc/os-release
```

Output:

```
PRETTY_NAME="Kali GNU/Linux  
Rolling" NAME="Kali  
GNU/Linux"  
VERSION_ID="2026.x"  
ID=kali  
ID_LIKE=Debian
```

2.2 Check Kernel

```
uname -a
```

2.3 Check Current User

```
Whoami
```

Output:

```
Student
```

2.4 Check IP Configuration

```
ip addr
```

```
# or
```

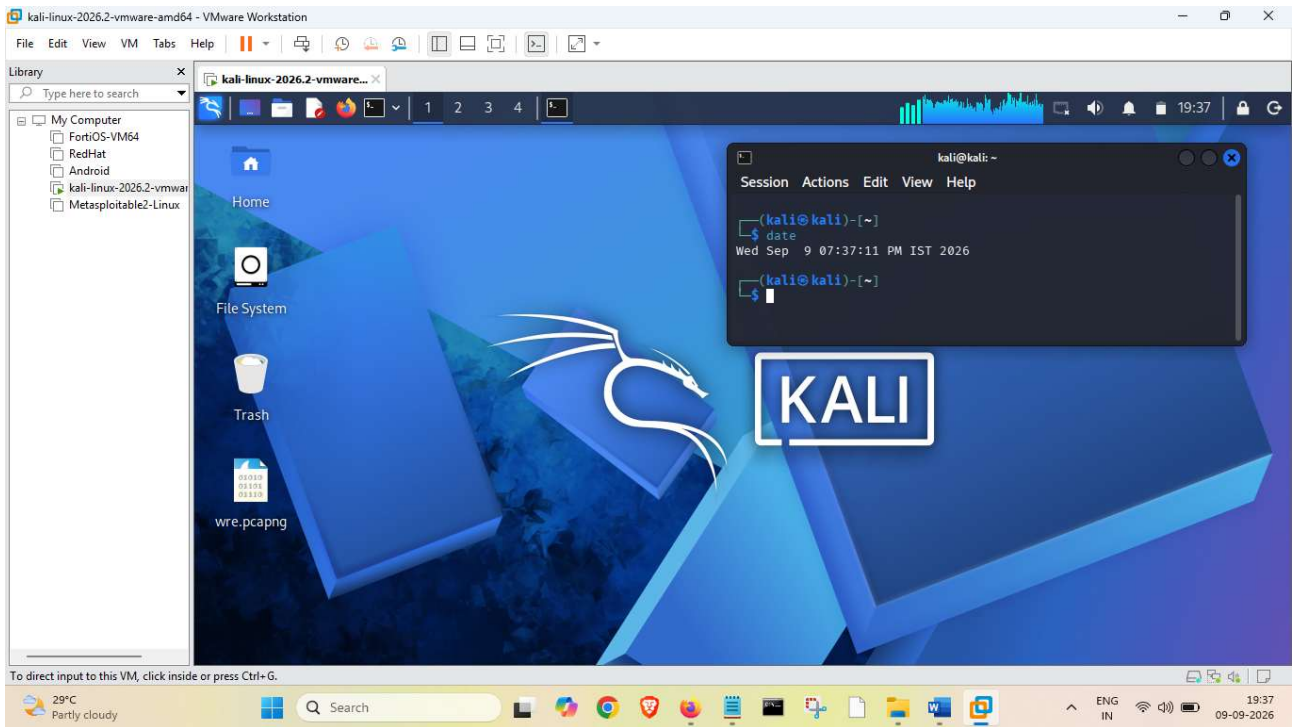
```
ip a
```

Lab address used in this report: 192.168.23.129/24

Screenshot Evidence

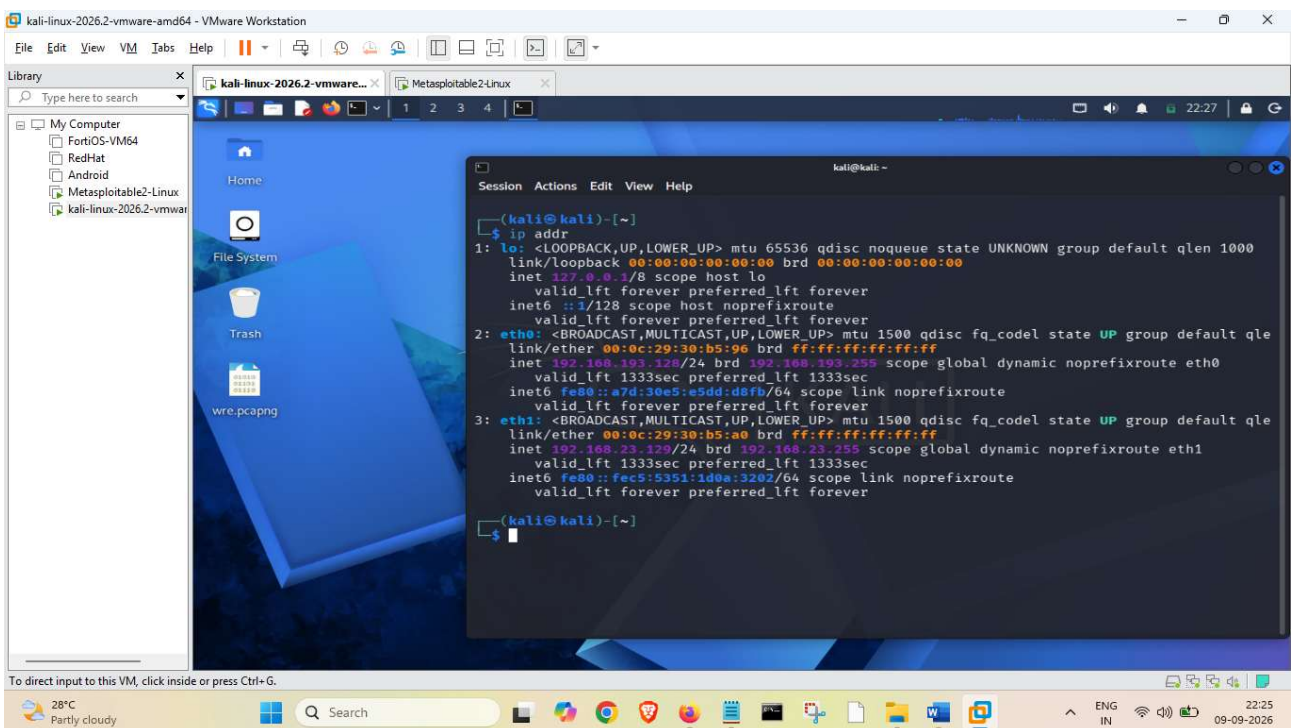
Screenshot 1 — Kali Linux Desktop

Evidence placeholder: Screenshot showing the Kali desktop, terminal, and date/time.



Screenshot 2 — Kali IP Configuration

Evidence placeholder: Terminal screenshot showing the output of **ip addr** and the lab IP address.



3. Step 2 — Configure Metasploitable 2

Start the Metasploitable 2 VM. After login, identify the network interface and verify the routing configuration.

3.1 Identify Network Interface

```
ifconfig  
# or  
ip addr
```

Lab address used in this report: 192.168.23.128/24

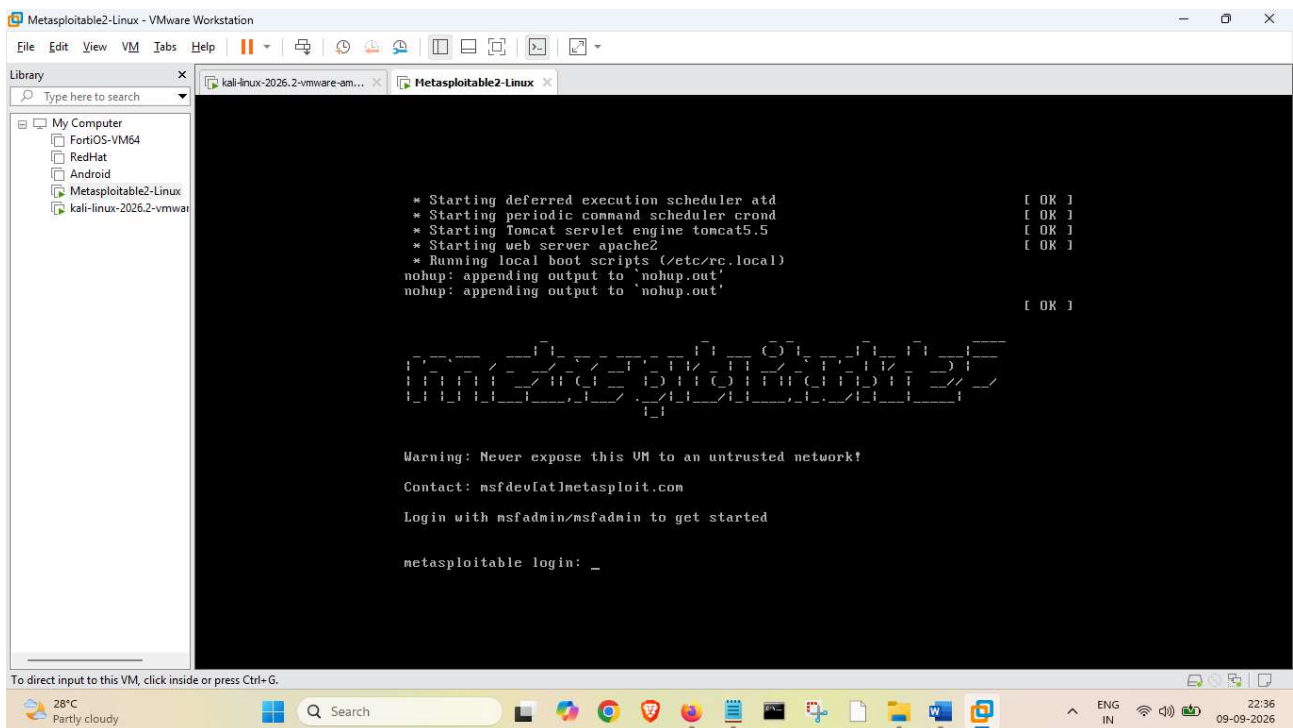
3.2 Check Routing

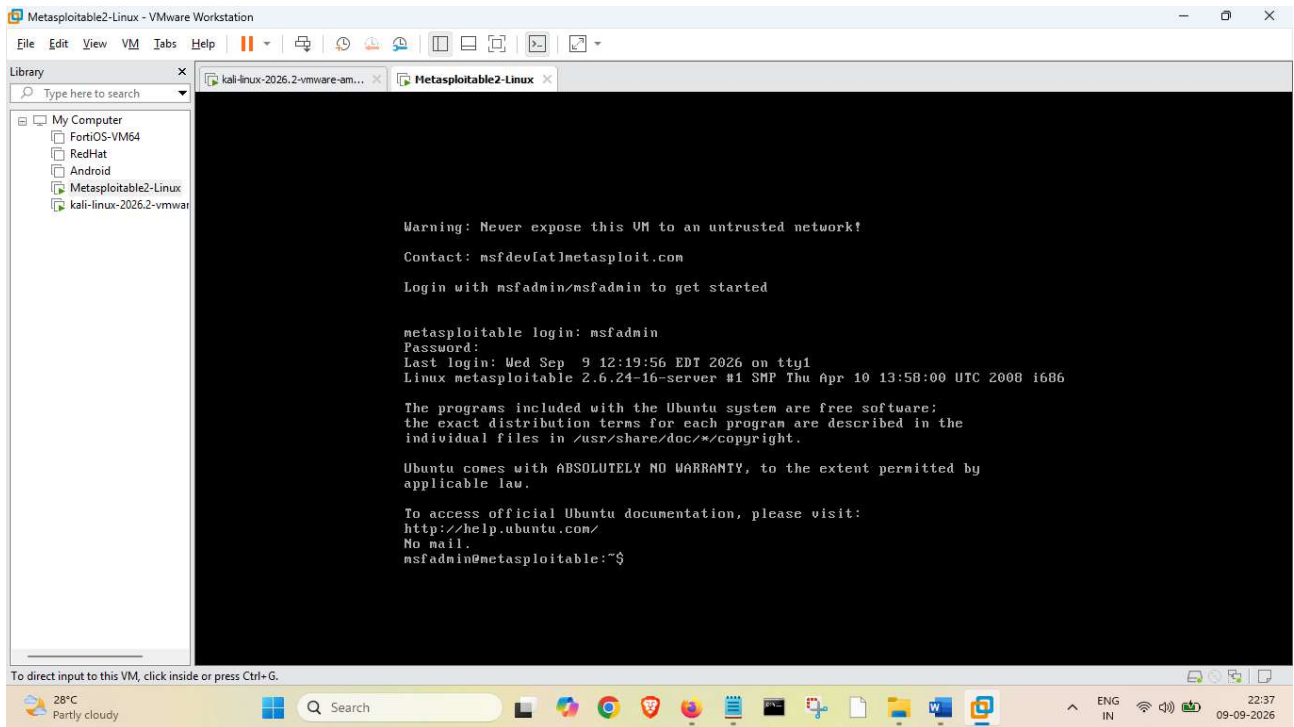
```
route -n  
  
# or  
  
ip route
```

Screenshot Evidence

Screenshot 3 — Metasploitable Login

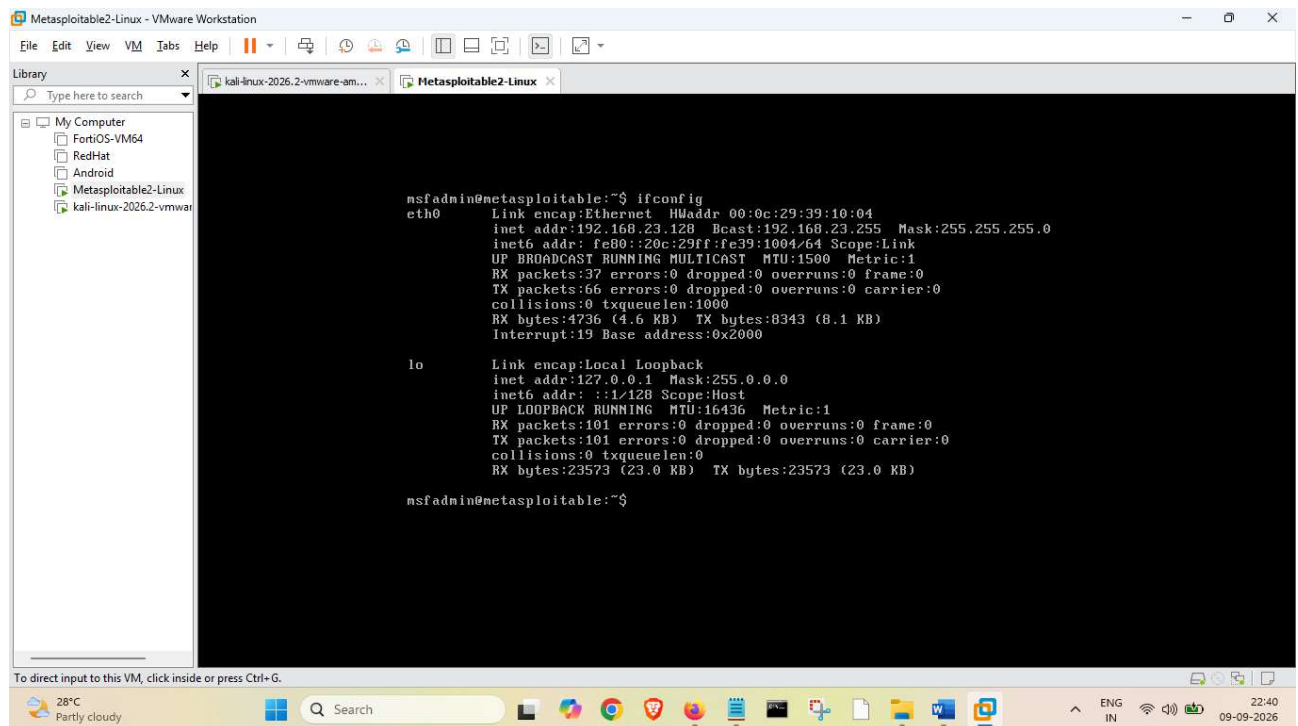
Evidence placeholder: Screenshot of the Metasploitable console after successful login.





Screenshot 4 — Metasploitable Network Configuration

Evidence placeholder: Output of `ifconfig` or `ip addr` showing the assigned address.



4. Step 3 — Verify Network Connectivity

From Kali Linux, send four ICMP echo requests to the Metasploitable 2 VM. A successful response confirms basic network-layer connectivity between the two laboratory systems.

4.1 Ping Test

```
ping -c 7 192.168.23.128
```

Output:

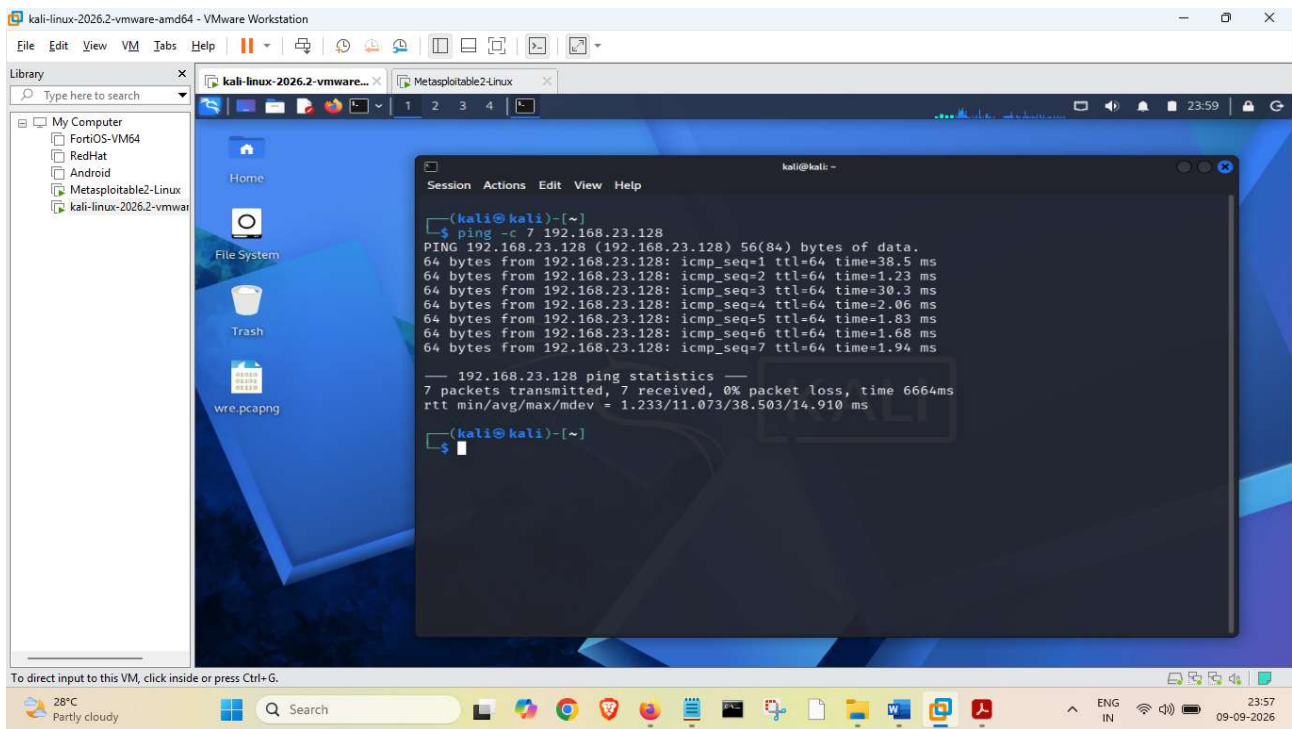
```
PING 192.168.23.128 (192.168.23.128) 56(84) bytes of data.  
64 bytes from 192.168.23.128: icmp_seq=1 ttl=64 time=38.5 ms  
64 bytes from 192.168.23.128: icmp_seq=2 ttl=64 time=1.23 ms  
64 bytes from 192.168.23.128: icmp_seq=3 ttl=64 time=30.3 ms  
64 bytes from 192.168.23.128: icmp_seq=4 ttl=64 time=2.06 ms  
64 bytes from 192.168.23.128: icmp_seq=5 ttl=64 time=1.83 ms  
64 bytes from 192.168.23.128: icmp_seq=6 ttl=64 time=1.68 ms  
64 bytes from 192.168.23.128: icmp_seq=7 ttl=64 time=1.94 ms  
  
--- 192.168.23.128 ping statistics ---  
7 packets transmitted, 7 received, 0% packet loss, time 6664ms  
rtt min/avg/max/mdev = 1.233/11.073/38.503/14.910 ms
```

Interpretation

The sample result shows seven replies and 0% packet loss. This indicates that Kali can successfully reach Metasploitable 2 over the isolated virtual network.

Screenshot 5 — Kali → Metasploitable Ping

Evidence placeholder: Screenshot showing the Kali terminal, destination IP, successful replies, and packet-loss statistics.



5. Step 4 — Wireshark Test Capture

Launch Wireshark on Kali Linux, select the interface connected to the isolated lab network, and start a packet capture.

```
wireshark
```

Generate ICMP traffic from Kali:

```
ping -c 7 192.168.23.128
```

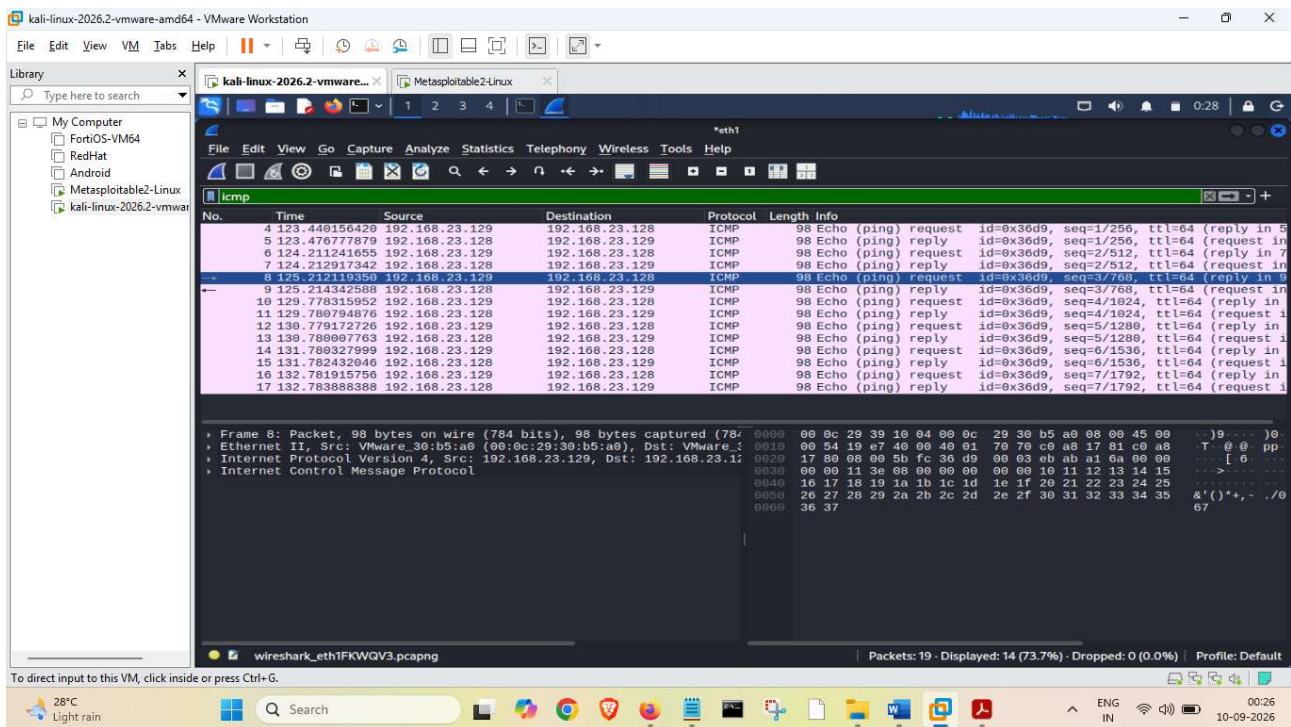
Apply the Wireshark display filter:

```
icmp
```

The capture should contain ICMP Echo Request and Echo Reply packets. Stop the capture and save it as **task1_wireshark_test_capture.pcapng**.

Screenshot 6 — Wireshark ICMP Capture

Evidence placeholder: Screenshot showing the ICMP filter and captured packets.



6. Conclusion

The sample laboratory configuration demonstrates the essential setup and verification stages for an isolated cybersecurity practice environment. Kali Linux provides the security-testing workstation, Metasploitable 2 provides a deliberately vulnerable target, and Wireshark provides packet-level visibility. Successful connectivity testing establishes a foundation for later reconnaissance, vulnerability assessment, exploitation, and incident-response exercises.